What you need to know

1. Mutually exclusive events

If A and B are mutually exclusive events, then

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B).$$

In general, if A and B are any 2 events, not necessarily mutually exclusive, then

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

2. Independent events

Two events A and B are said to be *independent* if and only if

$$P(A \text{ and } B) = P(A \cap B) = P(A)P(B).$$

3. Conditional probability of A given B, $P(A|B)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

4. Partitions, Total probability, Bayes theorem

$\{B_1, B_2, \cdots, B_k\}$ is a *partition* of S if

(a) $B_1, B_2, \cdots, B_k$ are disjoint subsets of S, and

(b) $B_1 \cup B_2 \cup \cdots \cup B_k = S \quad (\sum_i P(B_i) = 1)$

Let A be an event of S. Then,

$$P(A) = \sum_{j=1}^{k} P(A|B_j)P(B_j).$$

For $r = 1, 2, \cdots, k$,

$$P(B_r|A) = \frac{P(A|B_r)P(B_r)}{\sum_{j=1}^{k} P(A|B_j)P(B_j)}.$$

5. Random variable X

(a) X is discrete

Probability mass function $P(k)$

$$\sum_{k} P(k) = 1$$

(b) X is continuous

Probability density function $f(x)$

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$P(a < X < b) = P(a \leq X \leq b) = \int_{a}^{b} f(x)dx = F(b) - F(a)$$

6. The *distribution function* of a r.v. X (discrete or continuous) is the function

$$F(x) = P(X \leq x)$$

$$= \begin{cases} \int_{-\infty}^{x} f(y)dy, & X \text{ is continuous} \\ \sum_{k=-\infty}^{x} P(X = k), & X \text{ is discrete} \end{cases}$$

7. Mean and variance of X

$$E(X) = \begin{cases} \sum_x xP(x), & X \text{ is discrete} \\ \int_{-\infty}^{\infty} xf(x)dx, & X \text{ is continuous.} \end{cases}$$

In general,

$$E(g(X)) = \begin{cases} \sum_x g(x)P(x), & X \text{ is discrete} \\ \int_{-\infty}^{\infty} g(x)f(x)dx, & X \text{ is continuous.} \end{cases}$$

$Var(X) = E(X^2) - [E(X)]^2$

The *standard deviation* of a r.v. X is $\sigma_X = \sqrt{Var(X)}$.

Suppose a and b are constants, we have

$$E(aX+b) = aE(X)+b, \qquad Var(aX+b) = a^2 Var(X).$$

8. Normal distribution (continuous)

Suppose $X \sim N(\mu, \sigma^2)$. Then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

$\phi(z) = P(Z \leq z)$ **is tabulated.**

9. Poisson distribution (discrete)

X is the no. of random arrivals of events within a period of time or space.

λ is the average no. of random arrivals of events within a period of time or space.

$X \sim P(\lambda)$

$$P(X = k) = \frac{e^{-\lambda}\lambda^{k}}{k!}, \qquad k = 0, 1, 2, \cdots, \infty.$$

10. Binomial distribution (discrete)

X is the no. of successes observed in a binomial experiment.

The binomial experiment has n trials, and the probability of success of a single trial is p.

$X \sim B(n, p)$

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \cdots, n.$$

11. Using normal distribution to approximate binomial distribution

$X \sim B(n, p)$ **with** $np \geq 5$ **and** $n(1-p) \geq 5$

$\Rightarrow$ X **is approximately** $N(np, np(1-p))$.

E N V E L O P !

$$P_b(X = k) \approx P_n(k - 0.5 \leq X \leq k + 0.5)$$

$$P_b(a \leq X \leq b) \approx P_n(a - 0.5 \leq X \leq b + 0.5)$$

$$P_b(X \leq b) \approx P_n(X \leq b + 0.5)$$

$$P_b(X \geq a) \approx P_n(X \geq a - 0.5)$$

$$P_b(X < b) = P_b(X \leq b - 1) \approx P_n(X \leq b - 1 + 0.5)$$

$$P_b(X > a) = P_b(X \geq a + 1) \approx P_n(X \geq a + 1 - 0.5)$$

12. Sample mean $\overline{X}$

(a) $E(\overline{X}) = \mu, \quad Var(\overline{X}) = \dfrac{\sigma^2}{n}, \quad \sigma_{\overline{X}} = \dfrac{\sigma}{\sqrt{n}}$

(b) $X \sim N(\mu, \sigma^2) \Rightarrow \overline{X} \sim N\left(\mu, \dfrac{\sigma^2}{n}\right)$ — this holds for any n

(c) Central Limit Theorem

$$n \geq 30 \Rightarrow \overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$ — this holds for any X

13. Sample variance S^2

Theorem on p. 214

$$X \sim N(\mu, \sigma^2) \Rightarrow \frac{(n-1)S^2}{\sigma^2} \sim \text{chi-square with } (n-1) \text{ deg of freedom}$$

Confidence interval

μ

σ known

- $\bar{X} \sim N$
- use Z
- $[\bar{x} - E, \bar{x} + E]$
- $E = z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}$

σ unknown

- $\bar{X} \sim N$
- use t, $(n-1)$ d.f.
- $[\bar{x} - E, \bar{x} + E]$
- $E = t_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$

σ^2

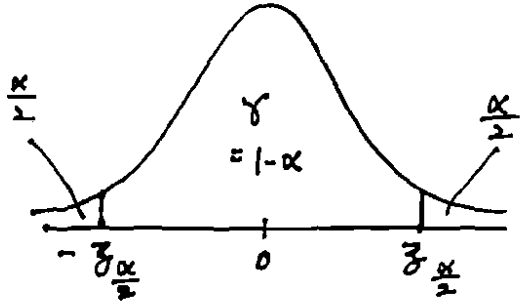
- $X \sim N$
- use χ^2
- $\left[\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2}}}, \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2}}} \right]$

γ% confidence interval for μ

When σ is known

require : $\bar{X}$ is approx. normal

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$



$$\left[\bar{x} - E,\ \bar{x} + E\right] \text{ where}$$

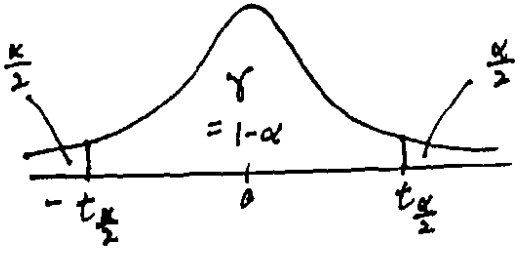
$$E = z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}$$

γ% confidence interval for μ

When σ is unknown

require : $\bar{X}$ is approx normal

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t \text{ distribution with } (n-1) \text{ d.f.}$$



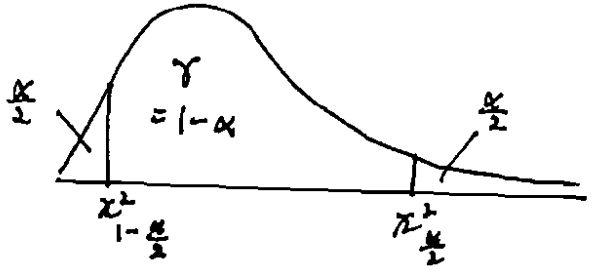
$$\left[\bar{x} - E,\ \bar{x} + E\right] \text{ where}$$

$$E = t_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$$

γ confidence interval for σ^2

require : X is approx normal

$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2$ distribution with $(n-1)$ d.f.



$$\left[\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2}}} , \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}}} \right]$$

Hypothesis testing

4 steps :

(I) Write down H_0 (always '=') & H_1 ($<$ / $>$ / $\neq$)

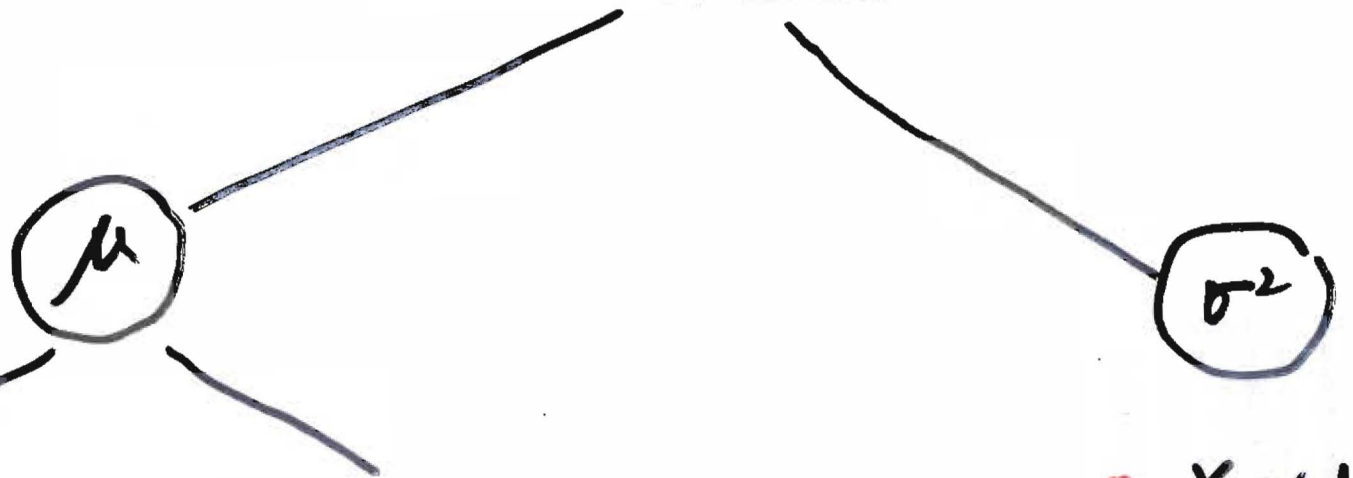
can swap:

(II) Calculate test statistic

(III) Draw rejection region

(IV) Conclude : accept ?, reject ?
Write in words about H_1
answer to the question

Hypothesis testing



σ known

- $\bar{X} \sim N$
- $Z = \dfrac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$

σ unknown

- $\bar{X} \sim N$
- $t = \dfrac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$, $(n-1)$ d.f.

σ^2

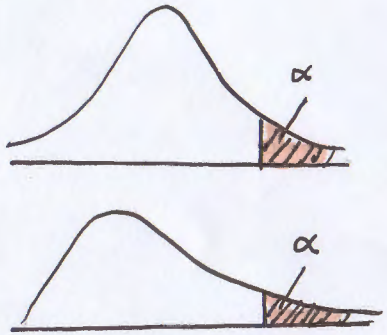
- $X \sim N$
- $\chi^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$,

  $(n-1)$ d.f.

Rejection (critical) region

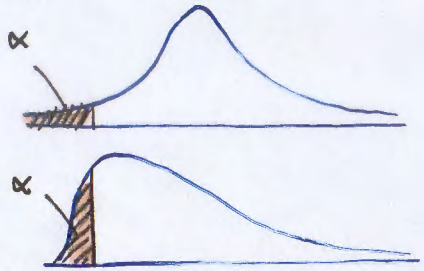
Upper tail test

$$\begin{cases} H_1: \mu > \mu_0 \\ H_1: \sigma^2 > \sigma_0^2 \end{cases}$$



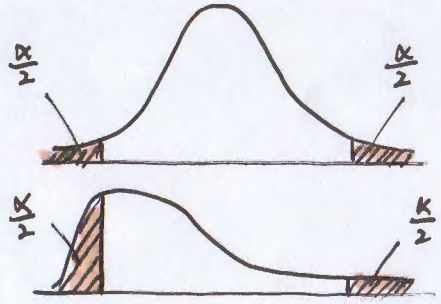
Lower tail test

$$\begin{cases} H_1: \mu < \mu_0 \\ H_1: \sigma^2 < \sigma_0^2 \end{cases}$$



Two tail test

$$\begin{cases} H_1: \mu \neq \mu_0 \\ H_1: \sigma^2 \neq \sigma_0^2 \end{cases}$$



Test concerning μ

When σ is known

- require : $\bar{X}$ is approx normal

$$\boxed{H_0 : \mu = \mu_0}$$

- test statistic :

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

- Critical value c :

lower tail test $P(Z < c) = \alpha$
upper tail test $P(Z > c) = \alpha$

two-tail test

$$P(Z < c_1) = P(Z > c_2) = \frac{\alpha}{2}$$

- If value of test statistic falls within the critical region (rejection region),

reject H_0 & accept H_1

Otherwise,

accept H_0 & reject H_1

Test concerning μ

When σ is known

- require : $\bar{X}$ is approx normal

$$H_0 : \mu = \mu_0$$

- test statistic :

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

- Critical value c :

lower tail test $P(Z < c) = \alpha$

upper tail test $P(Z > c) = \alpha$

two-tail test $P(Z < c_1) = P(Z > c_2) = \frac{\alpha}{2}$

Test concerning μ

When σ is unknown

- require : $\bar{X}$ is approx normal

$$H_0 : \mu = \mu_0$$

- test statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad d.f. = n-1$$

- Critical value c :

lower tail test $P(t < c) = \alpha$

upper tail test $P(t > c) = \alpha$

two-tail test $P(t < c_1) = P(t > c_2) = \frac{\alpha}{2}$

Test concerning σ^2

- require : X is normal

H_0 : $\sigma^2 = \sigma_0^2$

- test statistic :

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}, \quad \text{d.f.} = n-1$$

- Critical value c

lower tail test $P(\chi^2 < c) = \alpha$
upper tail test $P(\chi^2 > c) = \alpha$
two tail test $P(\chi^2 < c_1) = P(\chi^2 > c_2) = \frac{\alpha}{2}$